

DATE
19 March
Tuesday!

BOILER: It is defined as a closed vessel in which steam is produced from water by combustion of fuel.

The steam generated is employed for the following purposes.

- (i) For generating power in steam engine or steam turbines
- (ii) In textile industries for sizing and bleaching etc.
In other industries like sugar mills, chemical industries.
- (iii) For heating the buildings in cold weather and for producing hot water for hot water supply.

Primary Requirement of Steam generator or Boilers:-

- (1) The water must be contained safely.
- (2) The steam must be safely delivered in desired condition (as regards its pressure, temperature, quality and required rate).
- (3) All precautions must be applied safely.

CLASSIFICATION OF BOILERS →

- The boilers may be classified as follows:
- 1) Horizontal, Vertical or Inclined. [On the basis of axis position]
 - 2) Fire Tube and Water Tube.
In fire tube boilers, the hot gases are inside the tubes and the water surrounds

The tubes.

In the water tube boilers, the water is inside the tubes and hot gases surround them.

3. On the basis of shell type :-

4. Externally fired and Internally fired
→ The boiler is known as externally fired if the fire is outside the shell.
On the basis of Position → The boiler is known as internally fired if the furnace is located inside the boiler shell.

5. Forced Circulation and Natural Circulation
In forced circulation type of boilers, the circulation of water is done by a forced pump.

In natural circulation type of boilers, circulation of water in the boiler takes place due to natural convection currents produced by the application of heat.

(3.5-10 bar)

(10-25)

6. High pressure and low pressure boiler/Medium
→ The boilers which produce steam at pressure of 30 bar and above are called high pressure boiler.

→ The boilers which produce steam at pressure below 80 bar are called low pressure boilers.

7. On the Basis of Mobility / Stationary
→ Stationary boilers → are used for Power plant steam, for plant process steam etc.

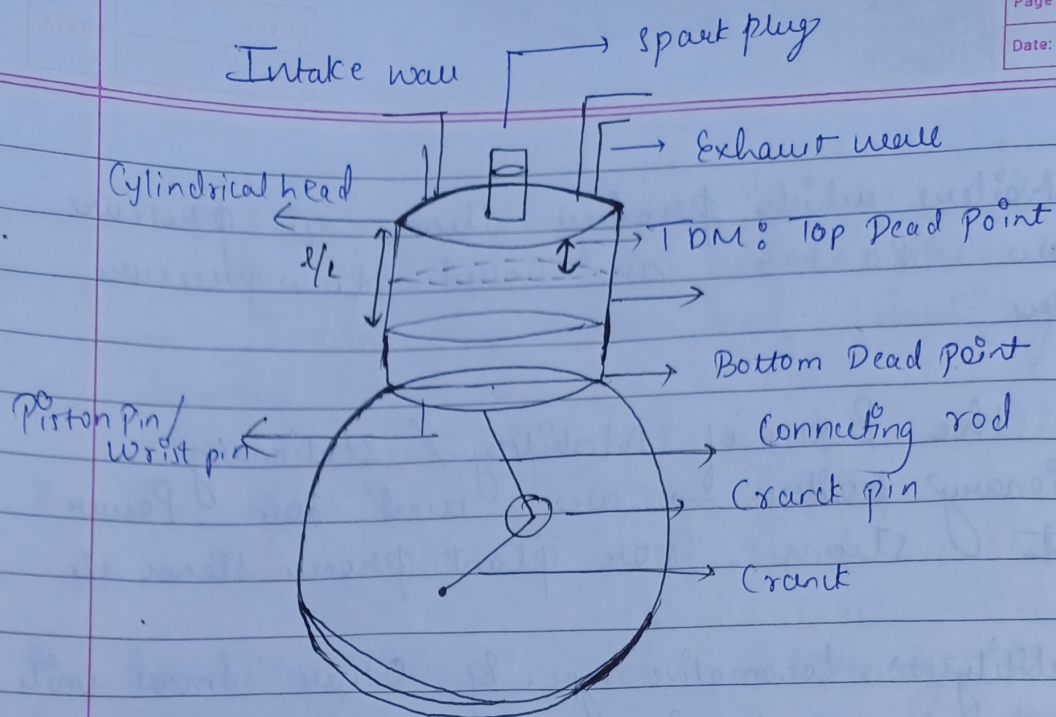
→ Mobility → locomotive type, & other small unit for temporary use at sites.

8. Single Tube & Multi-Tube Boilers

9. On the Basis of fuel:
Solid / Liquid / Gas / Nuclear

* A Furnace Grate is provided in a SB for supporting solid fuel in the furnace.
why: → To allow fresh air for the combustion to the fuel.

To Remove waste material / ash from the furnace / fuel so that combustion takes place properly.



$$V_{cc} = 10^{-3} \text{ l}$$

Boiler Mounting And Accessories!

Boiler Mountings:-

These are different fittings and devices which are necessary for the operation and safety of a boiler. Usually these devices are mounted over boiler shell.

Boiler Accessories:-

These are auxiliary plants required for steam boilers for their proper operations and for the increase of their efficiency.

Boiler Mountings:-

(1) Water level indicator:-
The function of a water level indicator is to indicate the level of water in the boiler constantly. It is also called water gauge.

(2) Pressure Gauge

The function of a pressure gauge is to measure the pressure exerted inside the vessel.

(3) Safety Valve

The function of a safety valve is to release the excess steam when the pressure of steam inside the boiler exceeds the rated pressure.

(4) Feed check Valve

To prevent the backflow of water from boiler to pump when $P_B > P_A$

→ TYPES:-

- 1) Dead weight Safety Valve.
- 2) Lever Safety Valve
- 3) Spring loaded Safety Valve
- 4) High steam and low water Safety Valve.

Feed check Valve:-

The function of a feed check valve is to control the supply of water to the boiler and to prevent the escaping of water from the boiler when the pump pressure is low.

Or Pump is stopped.

(5)

Stop Valve :- / Junction
A junction valve is a valve which is placed directly over a boiler and connected to a steam pipe which carries steam to the engine. If a valve is placed in the steam pipe leading steam to the engine and placed near the engine, it is usually termed as stop valve.

Boiler accessories :-

① All the devices and plants that are installed with boiler to improve efficiency!

1) Economiser :-

An Economiser is a device in which the waste heat of the flue gases is utilised for heating the feed water.

2) Man holes :-

Access to underground piping, meters and facilities.

3) Blew of Cock :-

To discharge mud and other sediments deposited in the bottom most part of the water space in the boiler, while boiler is in operation.

* $C_{p, \text{air}}$ Preheat heat
 Latent heat of vaporization = 0

4) Air Preheater
 The function of the air pre-heater is to increase the temperature of air before it enters the furnace. It is generally placed after the Economiser so the flue gases pass through the Economiser & then to the air preheater.

NOTE: Storage type air preheaters are employed widely in larger plants.

a) A Boiler produces 8 kg of steam per kg of coal at the pressure of 1 MPascal at feed water temperature 40°C . Equivalent temperature, Calculate the Equivalent Evaporation Assuming Dryness factor of 0.95. Also find the Efficiency of boiler if Calorific value is 32 MJ/kg .

Ans - Given that :-

$$[1 \text{ mPa} = 10^6 \text{ Pascal}]$$

$$M_s = 8 \text{ kg}$$

$$[1 \text{ bar} = 10^5 \text{ Pascal}]$$

$$M_g = 1 \text{ kg}$$

$$P_b = \text{mPa} = 10 \text{ bar}$$

$$T_{\text{feed}} = 40^\circ\text{C}$$

$$h_f = 167.6 \text{ J/kg}$$

$$x = 0.95$$

$$C.V = 32 \text{ MJ/kg}$$

$$\eta =$$

In a boiler trial 1200 bare kg of Coal
 is produced in 24 hour duration of a
 boiler generates 11500 kg of dry
 saturated steam at a pressure of 0.8
 Mega Pascal gauge at a feed water
 of 35°C . Calculate the
 Equivalent of fuel is Evaporation Per kg
 of Coal & Efficiency $C_{\text{ev}} \rightarrow 30000 \text{ kg/kg}$
 Specific heat of $\text{H}_2\text{O} \rightarrow 4.18 \text{ kJ/kgK}$

- a. A Boiler produces 8 kg Steam per kg of fuel at a Pressure of 11 bar from the feed water temperature of 75°C. The efficiency of boiler is 80 percent and the factor of utilization is 1.13. Determine
- 1) Temperature of Steam & Degree of Super heat
 - 2) Calorific value of Coal
 - 3) Equivalent Evaporation of per kg Coal

cp of Steam = 2.2 kJ/kgK

→ Given that,

$m_s = 8 \text{ kg}$

$m_o = 1 \text{ kg}$

$P_b = m P_a = 10 \text{ bar}$

find?

$\eta = 0.95$

i) $m_c = ?$

$CV = 32 \text{ mg/kg}$

2) $h_b = ?$

$h_f = 167.6 \text{ kJ/kg}$

$= 7628 + 0.95 \times 2015.3$

ii) $h_b = ?$

$$h_b = \frac{m_s (h - h_f)}{m_f \times C.V.}$$

$h_b = ?$

c) To provide a natural draught boiler & Equiv with height = 25 m $T_o = 30^\circ\text{C}$, $T_g = 300^\circ\text{C}$ If the amount of fuel supplied $m_a = 20^\circ\text{g}$.

- Theoretically Draught of H_2O .
- The Velocity of fuel gases leaving the chimney.
- Draught of Pascal in terms of mm.

$$(i) \Delta P = 353 \times 10 \times 25 \left[\frac{1}{303} - \frac{(20+1)}{20 \times 573} \right]$$

$$= 129.5394 \text{ Pa}$$

for mm of water Column: $1 \text{ Pa} \rightarrow 0.10 \text{ mm H}_2\text{O}$

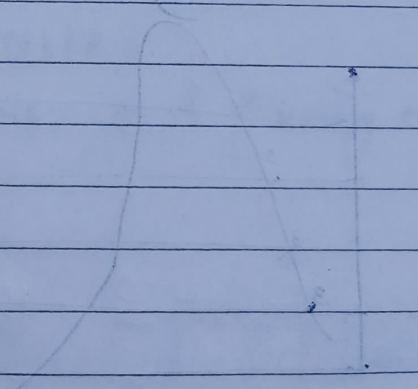
$$\Delta P = 129.5394 \times 0.10 \text{ mm H}_2\text{O}$$

$$= 1.295394 \text{ mm of H}_2\text{O}$$

49 50 51 52 53 54 55 56 57 58 59 60

11

(Em) V

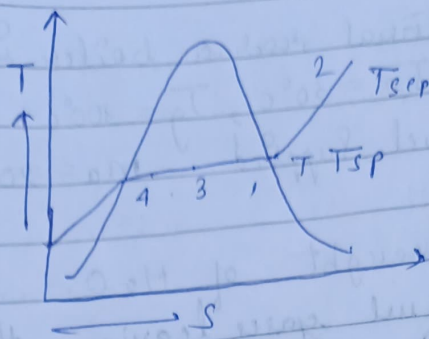


vertical draught of steam ← with boiler vessel &

air supply of fuel between itself

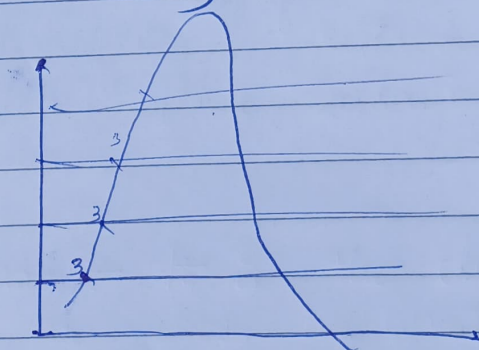
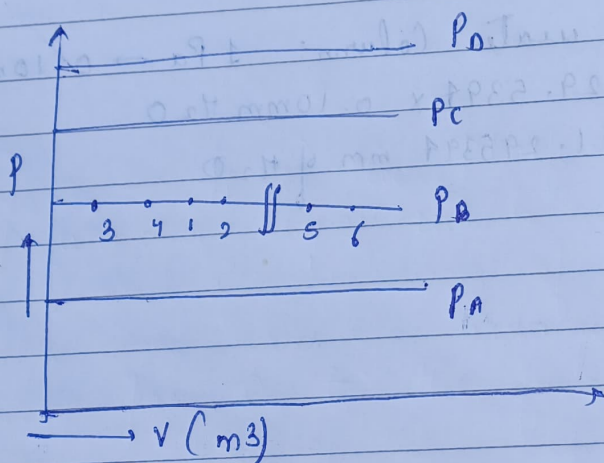
with steam & $0 < x < 1$ (ratio of weight)

with fuel ← 0 < 1



$$\int_1^2 ds = \int_1^2 \frac{dQ}{T}$$

$$s_2 - s_1 = \int_1^2 \frac{m C_p dT}{T}$$



* Super heated steam $\rightarrow$ H₂O ka temperature
 saturated heat se jyda ho.
 Dryness fraction x $0 < x < 1 \rightarrow$ wet steam
 $x = 1 \rightarrow$ Dry steam

COMPARISON BETWEEN WATER TUBE AND FIRE TUBE BOILER:

- | FIRE TUBE BOILER | WATER-TUBE BOILER |
|--|---|
| 1) Hot gases inside the tube and water outside the tube. | Water inside the tube and hot gases outside the tube. |
| 2) Generally internally fired | Externally fired |
| 3) Operating pressure limited to 16 bar | Can work under as high pressure as 100 bar. |
| 4) Lower rate of steam production | Higher rate of steam production |
| 5) Transportation is difficult | Transportation is simple. |
| 6) Construction is difficult | Construction is simple. |
| 7) chances of explosion are less | chances of explosion are more. |

SELECTION OF A BOILER:

- 1) The Working Pressure and Quality of Steam Required
- 2) Steam Generation rate.
- 3) Floor area available
- 4) Accessibility for repair and inspection.
- 5) Comparative initial cost.
- 6) The Portable load factor.
- 7) The fuel & water available
- 8) Operating and maintenance cost

Shell $\rightarrow$ The shell of a boiler consists of one or more steel plates bent into a cylindrical form and riveted or welded together. The shell ends are closed with the end plates.

Setting $\rightarrow$ The primary function of setting is to confine heat to the boiler and form a passage for gases. It is made of brickwork and may form the wall of the furnace of the combustion chamber. It also provides support in some type of boiler.

Grate $\rightarrow$ It is the platform in the furnace upon which fuel is burnt and it is made of cast iron bars. The bars are so arranged that air may pass on to the fuel for combustion. The area of the grate on which the fire rests in a coal or wood fired boiler is called grate furnace.

Furnace $\rightarrow$ It is the chamber formed by the space above the grate and below the boiler shell, in which combustion takes place. It is also called a fire-box.

Water space and steam space $\rightarrow$
 The volume of the shell that is occupied by the water is termed as water space while the entire shell volume less than the water and tubes space is called steam space.

Water level :-

The level at which water stands in the boiler is called water level. The space above the water level is called steam space.

Foaming :- formation of steam bubbles on the surface of boiler water due to high surface tension of the water.

Scale :- A deposit of medium to extreme hardness occurring on water heating surface of a boiler because of an undesirable condition in the boiler water.

Blowing off :- The removal of the mud and other impurities of water from the lowest part of the boiler (where they actually settle) is termed as 'blowing off'. This is accomplished with the help of a blow off cock or valve.